

Adaptive Transfer Learning with EfficientNetB1 for Multi-Class Brain Tumor Identification in MRI Images

Abstract—Deep convolutional networks can reach high accuracy on public brain magnetic resonance imaging (MRI) datasets, but the reported numbers are strongly affected by data version, partitioning, preprocessing, and fine-tuning choices. This work evaluates a controlled EfficientNetB1 transfer-learning pipeline on the current balanced 7,200-image Brain Tumor MRI dataset containing glioma, meningioma, pituitary tumor, and no-tumor images. The original training and testing folders were first merged class-wise, after which a reproducible random image-level split with seed 42 assigned 250 images per class to a held-out test set. The remaining 1,550 images per class formed the development pool; a 10% validation split produced 5,580 training and 620 validation images. Images were foreground-cropped and resized to 240×240 pixels. EfficientNetB1 was initialized with ImageNet weights, earlier backbone stages were frozen, and only block6, block7, and the top convolution were fine-tuned while backbone batch-normalization layers remained frozen. On the 1,000-image test set, the model achieved 98.00% accuracy and a macro F1-score of 98.00%. Class recall was 96.0% for glioma, 98.0% for meningioma, 98.8% for no-tumor, and 99.2% for pituitary tumor. Error analysis showed that the largest residual confusion was between glioma and meningioma. The results demonstrate strong image-level performance while also highlighting the need for patient-level and external validation before clinical interpretation.

Index Terms—brain tumor classification, magnetic resonance imaging, EfficientNetB1, transfer learning, fine-tuning, deep learning, medical image classification

I. INTRODUCTION

Brain tumors exhibit substantial variation in location, appearance, size, and tissue composition, making reliable classification from magnetic resonance imaging (MRI) a difficult image-analysis problem. Automated classification is therefore commonly investigated as a computer-aided analysis task rather than as a replacement for radiological diagnosis. Convolutional neural networks (CNNs) are particularly attractive because they learn hierarchical visual representations directly from images, but training a deep model from scratch is often impractical when labeled medical data are limited. Transfer learning offers a practical alternative by adapting representations learned from a large source dataset to a medical target task [1], [2].

EfficientNet is well suited to this setting because its compound-scaling strategy jointly balances network depth, width, and input resolution instead of increasing a single dimension in isolation [3]. The family also transfers effectively from ImageNet pretraining [4]. However, performance reported for brain tumor MRI classification is not determined

by architecture alone. Dataset revisions, train–test partitions, image augmentation, resizing, and the amount of backbone fine-tuning can all materially change the measured result. This is especially relevant for the widely used Brain Tumor MRI Dataset hosted on Kaggle, whose current release contains 7,200 images evenly divided across four classes [5].

Recent work on the same four-class task has used hybrid, attention-based, and ensemble designs. Ahmad *et al.* combined MobileNetV2 and SE-ResNet50 in SEMoNet and reported 93.2% accuracy on 7,200 images [6]. Uthra and Sivashankari proposed a query-guided cross-attention CNN with ensemble learning and reported 95.50% held-out test accuracy [7]. Alshammari and Mahmood evaluated a multi-scale EfficientNetV2-S framework on the same 7,200-image four-class dataset as an external related task and obtained 95.00% accuracy [8]. Earlier ICCIT work by Sikder *et al.* reported 98.05% accuracy using a Dense-ResNet triple-level ensemble on a different three-class, 3,064-image contrast-enhanced MRI dataset [9]. These studies show that high accuracy is possible, but direct numerical comparison is unsafe when datasets and evaluation protocols differ.

Rather than proposing a new neural architecture, this study focuses on a reproducible evaluation pipeline for the revised balanced dataset. The main contributions are as follows:

- A class-wise merge and reproducible random re-splitting procedure is used to form a balanced 1,000-image held-out test set, with strict count verification before preprocessing and training.
- A foreground-focused crop and fixed 240×240 resizing stage is combined with mild training-only augmentation.
- EfficientNetB1 is selectively fine-tuned by updating block6, block7, and the top convolution while freezing earlier backbone layers and all backbone batch-normalization layers.
- Performance is examined at the class level using a confusion matrix, precision, recall, F1-score, and specificity, exposing the remaining glioma–meningioma confusion rather than reporting accuracy alone.

II. RELATED WORK

Transfer learning has been repeatedly shown to be useful in medical imaging when task-specific training data are smaller than the datasets used to pretrain modern CNNs. Tajbakhsh *et al.* found that adequately fine-tuned pretrained CNNs can match or outperform training from scratch across multiple

medical imaging applications [1]. For brain MRI, Swati *et al.* demonstrated that fine-tuning a pretrained deep CNN can provide strong tumor classification performance, reinforcing the feasibility of reusing natural-image representations for MRI analysis [2].

Early CNN-based studies established the effectiveness of automated brain tumor recognition from MRI images. Cheng *et al.* introduced a benchmark dataset and demonstrated that image-based features can be effectively used for multi-class brain tumor analysis [10]. Abiwinanda *et al.* evaluated a convolutional neural network approach for brain tumor classification and showed the potential of end-to-end feature learning for MRI images [11]. Sultan *et al.* proposed a CNN architecture for multi-class brain tumor classification and reported improved performance compared with conventional approaches [12]. Deepak and Ameer investigated transfer learning using pretrained CNN features and highlighted the advantage of feature reuse for brain tumor MRI classification tasks [13].

The choice of backbone and evaluation protocol remains important. EfficientNet introduced a systematic scaling rule that provides a strong accuracy–efficiency trade-off [3]. In more recent four-class brain tumor studies, increasingly complex frameworks have been explored. SEMoNet fused lightweight MobileNetV2 representations with squeeze-and-excitation ResNet features and achieved 93.2% accuracy on 7,200 MRI images [6]. QGC-CNN used learnable query tokens to model longer-range dependencies and combined several CNN predictors through a LightGBM stacking framework; the final reported test accuracy was 95.50% [7]. MCND-ComputeNet++ used multi-scale feature fusion, attention-based aggregation, calibration, and test-time augmentation; when evaluated on the 7,200-image four-class dataset as an external task, it achieved 95.00% accuracy and 94.90% macro F1-score [8].

A related ICCIT study proposed a Dense-ResNet ensemble with three attention mechanisms for glioma, meningioma, and pituitary classification and reported 98.05% accuracy on 3,064 T1-weighted contrast-enhanced MRI images [9]. That result is useful context but is not directly comparable with the present four-class experiment because both the dataset and label space differ. This distinction motivates the present work: the goal is not to claim universal superiority from a single accuracy value, but to document a controlled image-level experiment on the current balanced 7,200-image release.

III. DATASET AND DATA PREPARATION

A. Dataset

The experiments use the Brain Tumor MRI Dataset by Nickparvar [5]. The release used in this work contains 7,200 images in four classes: glioma, meningioma, no tumor, and pituitary tumor. The supplied directory structure contains 1,400 training and 400 testing images for each class, yielding 1,800 images per category.

Representative images are shown in Fig. 1. The collection contains axial, coronal, and sagittal views and visible variation

TABLE I
CLASS-WISE DATASET REARRANGEMENT USED IN THIS STUDY.

Class	Original Train	Original Test	Development Pool	Test Set
Glioma	1400	400	1550	250
Meningioma	1400	400	1550	250
No Tumor	1400	400	1550	250
Pituitary	1400	400	1550	250
Total	5600	1600	6200	1000

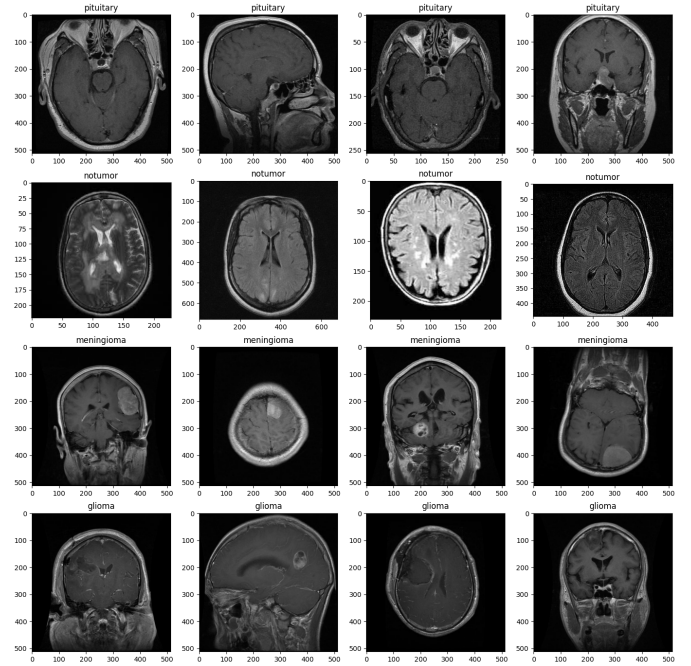


Fig. 1. Representative MRI samples from the four classes in the dataset.

in field of view, image dimensions, and contrast. Such heterogeneity is useful for training but also means that an image-level split cannot be interpreted as a patient-independent clinical evaluation.

B. Class-wise Merge and Re-splitting

The original training and testing folders were merged into a unified class-wise pool, with source prefixes added to filenames to prevent duplication. The merged dataset contained 1,800 images per class, resulting in 7,200 images. A stratified random split with seed 42 was then applied, assigning 250 images per class to the held-out test set and the remaining 1,550 images per class to the development pool. A further 10% validation split produced 5,580 training, 620 validation, and 1,000 test images. The split information was stored for reproducibility.

This experiment uses image-level splitting because patient identifiers are unavailable in the dataset. Therefore, possible subject-level correlation between partitions cannot be fully excluded.

C. Foreground Cropping and Resizing

To remove irrelevant background regions, MRI images were converted to grayscale and processed using Gaussian smoothing, thresholding, and morphological operations. The largest contour was used to obtain the foreground crop, while the original image was retained when no valid contour was detected. All images were resized to 240×240 pixels using the same deterministic preprocessing pipeline before model training and evaluation.

IV. METHODOLOGY

A. EfficientNetB1 Transfer Learning

EfficientNetB1 was initialized with ImageNet weights and the original classification top was removed. The input tensor had shape $240 \times 240 \times 3$. The backbone output was an $8 \times 8 \times 1280$ feature tensor, which was aggregated with global average pooling.

The fine-tuning policy was deliberately selective. Backbone layers belonging to block6, block7, and top_conv were marked trainable; earlier backbone stages remained frozen. All backbone batch-normalization layers were kept non-trainable, and the backbone was invoked with `training=False` so that their stored statistics were not updated during fine-tuning. This policy was chosen to adapt higher-level features while limiting unnecessary disturbance to earlier ImageNet representations.

The custom head consisted of a 512-unit ReLU dense layer with $L_2 = 10^{-4}$ regularization, batch normalization, and dropout of 0.30, followed by a 256-unit ReLU dense layer with the same L_2 penalty, batch normalization, and dropout of 0.20. A four-unit softmax layer produced class probabilities. The complete model contained 7,366,539 parameters, of which 6,798,728 were trainable in the implemented configuration.

B. Augmentation and Optimization

Training images were processed with EfficientNet input preprocessing and mild online augmentation: random rotation up to 5° , horizontal and vertical translation up to 5% of image size, and zoom up to 5%. Horizontal flipping was disabled because the dataset contains heterogeneous MRI orientations and unnecessary mirroring could alter anatomical laterality cues. Validation and test images were not augmented.

The model was optimized with Adam using an initial learning rate of 3×10^{-5} and categorical cross-entropy loss. Batch size was 32 and the maximum number of epochs was 50. Model checkpoints were selected by minimum validation loss. Early stopping used a patience of eight epochs and restored the best weights. ReduceLROnPlateau monitored validation loss with factor 0.3, patience 3, minimum improvement 10^{-4} , and minimum learning rate 10^{-7} . Training stopped after epoch 36 and restored the weights from epoch 28, which had the lowest validation loss.

C. Evaluation

The test generator was not shuffled, ensuring alignment between predictions and ground-truth labels. Accuracy, precision, recall, F1-score, specificity, and the confusion matrix were calculated from the 1,000 held-out images. For class c , precision and recall were calculated as

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c}, \quad (1)$$

with $F1_c = 2P_cR_c/(P_c + R_c)$. Specificity was computed as $TN_c/(TN_c + FP_c)$.

V. EXPERIMENTAL RESULTS

A. Training Behavior

The training and validation curves in Fig. 3 illustrate the optimization behavior of the proposed EfficientNetB1 model. Training accuracy increased rapidly during the initial epochs and gradually stabilized as learning progressed. Validation performance followed a similar trend, reaching approximately 97–98% accuracy without severe divergence between training and validation losses. The learning-rate scheduler and early stopping mechanism helped prevent unnecessary overfitting, with the best checkpoint obtained at epoch 28.

B. Held-out Test Performance

The final model achieved 98.00% accuracy with test loss 0.2111. The confusion matrix in Fig. 4 provides a class-wise view of prediction performance. Most samples were correctly classified, while the major remaining confusion occurred between glioma and meningioma categories, indicating the difficulty of separating visually similar tumor appearances.

C. ROC Curve Analysis

The receiver operating characteristic (ROC) curve provides an additional evaluation of the model's discriminative capability by illustrating the trade-off between the true positive rate and false positive rate at different classification thresholds. The multi-class ROC analysis of the proposed EfficientNetB1 model is presented in Fig. 5. The obtained curves demonstrate strong class-wise separability, indicating that the model can effectively distinguish among glioma, meningioma, pituitary tumor, and no tumor categories. The high area under the curve (AUC) values further confirm the robustness of the learned feature representations and the overall classification performance on the held-out test dataset.

Because the test set is perfectly balanced, macro recall is also 98.00%. Table II presents the class-wise results reconstructed from the confusion matrix.

Out of 1,000 test images, 980 were classified correctly. The most visible error pattern was glioma being predicted as meningioma: eight of the 250 glioma images fell into that category. One glioma image was predicted as no tumor and one as pituitary. Meningioma had five false negatives, while no-tumor and pituitary images were the most reliably recognized classes, with three and two false negatives respectively. The confusion pattern suggests that the principal residual challenge

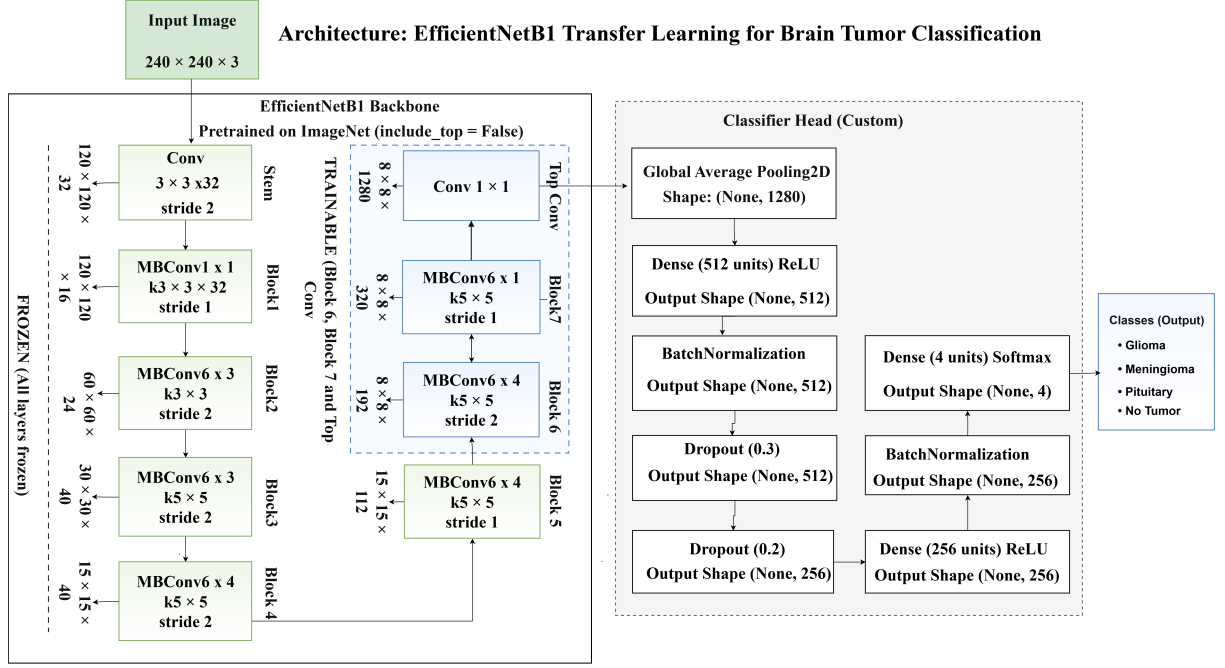


Fig. 2. Implemented EfficientNetB1 transfer-learning architecture for four-class brain tumor MRI classification.

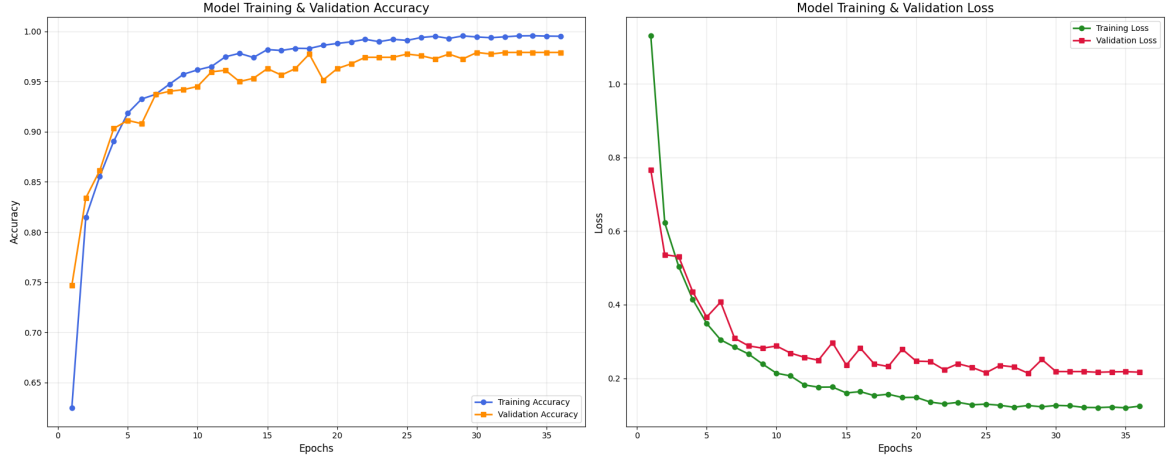


Fig. 3. Training and validation accuracy/loss curves of the proposed EfficientNetB1 model.

TABLE II
CLASS-WISE TEST PERFORMANCE (250 IMAGES PER CLASS).

Class	Precision	Recall	F1	Specificity
Glioma	98.36	96.00	97.17	99.47
Meningioma	95.33	98.00	96.65	98.40
No Tumor	99.20	98.80	99.00	99.73
Pituitary	99.20	99.20	99.20	99.73
Macro avg.	98.02	98.00	98.00	99.33

is separating glioma from meningioma rather than detecting the presence of an abnormality in general.

D. Comparison with Studies Using the Masoud Nickparvar 7,200-Image Dataset

Table III summarizes studies that evaluated brain tumor classification using the Masoud Nickparvar 7,200-image MRI dataset. Only results based on this dataset are included to maintain a fairer comparison. However, differences in data splitting, preprocessing, and evaluation protocols should still be considered when interpreting the results.

VI. DISCUSSION

Three observations are important. First, the balanced split makes overall accuracy easier to interpret because no class dominates the test set. Class-wise recall still matters: the glioma recall of 96.0% is lower than the other classes and

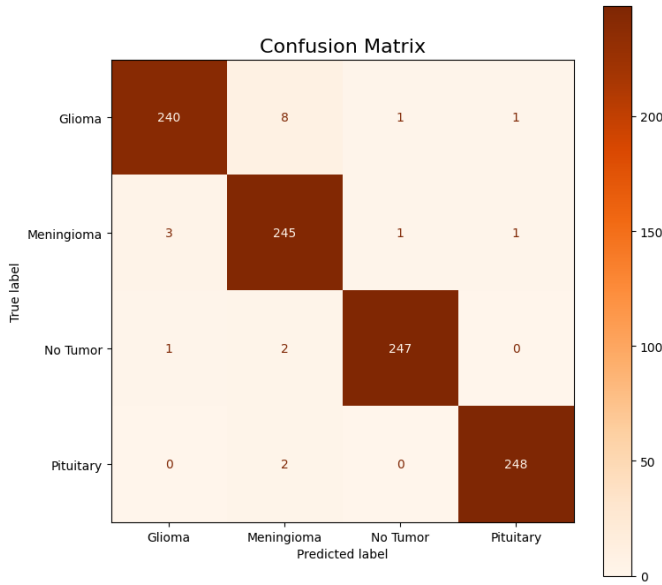


Fig. 4. Confusion matrix of the proposed model.

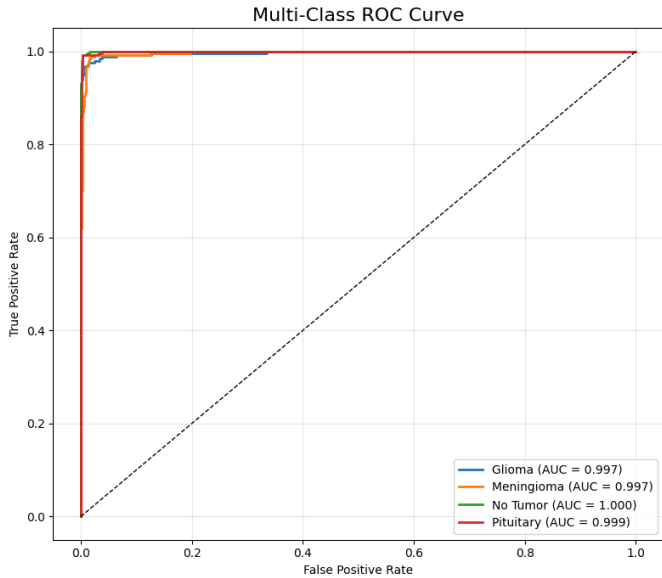


Fig. 5. Multi-class ROC curves of the proposed model.

accounts for half of all test errors. The eight glioma-to-meningioma errors are consistent with the visual challenge posed by heterogeneous tumor appearance and overlapping MRI characteristics.

Second, selective fine-tuning appears sufficient for strong performance. Updating the later EfficientNetB1 stages allows high-level features to adapt to tumor morphology, while freezing earlier stages and backbone batch-normalization statistics reduces the risk of destabilizing pretrained representations. The validation curve and the restored epoch-28 checkpoint suggest that the conservative learning rate and scheduler were effective for this dataset.

TABLE III
COMPARISON WITH STUDIES USING THE MASOUD NICKPARVAR 7,200-IMAGE DATASET.

Study	Model	Dataset	Accuracy
Ahmad <i>et al.</i> [6]	SEMoNet	Nickparvar (7,200)	93.20%
Uthra & Sivashankari [7]	QGC-CNN ensemble	Nickparvar (7,200)	95.50%
Alshammari & Mahmood [8]	MCND-ComputeNet++	Nickparvar (7,200)	95.00%
This work	EfficientNetB1 selective fine-tuning	Nickparvar v2 (7,200)	98.00%

Third, the data arrangement itself is part of the experimental definition. Merging the uploader’s original training and testing folders and generating a new image-level split answers a different question from evaluating on the original 1,600-image test folder. The advantage is a controlled, reproducible, perfectly balanced held-out set with identical per-class support. The disadvantage is that the result is no longer directly comparable to papers that preserve the uploader’s original split.

A. Threats to Validity

The most important limitation is the absence of patient identifiers. The random partition is performed at the image level, so slices originating from the same subject, or highly similar acquisitions, may appear in different subsets. Without patient-level metadata this risk cannot be eliminated. Consequently, the reported 98.00% accuracy should be interpreted as benchmark-level image classification performance rather than a patient-independent clinical estimate.

A second limitation is that evaluation is restricted to one public dataset and one random seed. Although the split is deterministic and reproducible, repeated splits or cross-validation would provide a more stable estimate of variance. Third, the work does not evaluate external datasets, scanner/domain shift, subgroup performance, probability calibration, or radiologist comparison. Finally, foreground cropping uses a fixed intensity threshold and largest-contour heuristic; while it worked for the evaluated images, the rule may not transfer unchanged to MRI data acquired under different protocols.

These limitations also define the most useful next steps: patient-grouped evaluation when identifiers are available, repeated stratified experiments, external validation, calibration analysis, uncertainty-based abstention, and explainability checks against radiologist-reviewed regions of interest.

VII. CONCLUSION

This study presented a reproducible EfficientNetB1 transfer-learning experiment for four-class brain tumor MRI classification on a balanced 7,200-image dataset. The original directory partitions were merged and randomly re-split class-wise, yielding 5,580 training, 620 validation, and 1,000 held-out test images. A foreground crop, fixed 240×240 resizing, mild augmentation, and selective fine-tuning of EfficientNetB1

block6, block7, and the top convolution produced 98.00% test accuracy and 98.00% macro F1-score. Pituitary and no-tumor images were classified most reliably, whereas glioma remained the most difficult class, with most errors directed toward meningioma. The result supports EfficientNetB1 as an effective image-level baseline for this balanced benchmark, but patient-level and external validation remain necessary before any clinical interpretation.

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